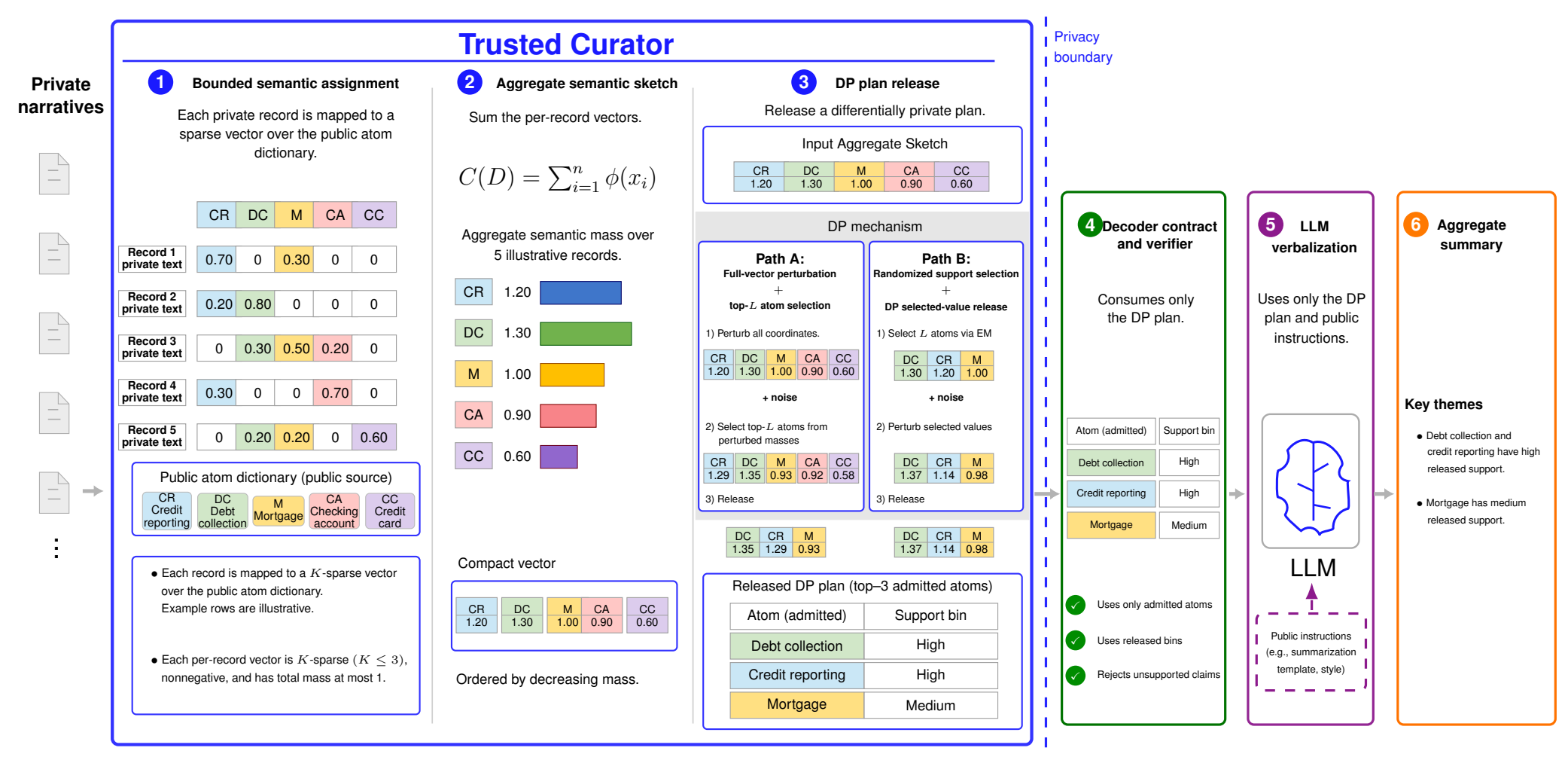

Private narratives
Trusted Curator
1 Bounded semantic assignment
Each private record is mapped to a sparse vector over the public atom dictionary.
CR DC M CA CC
Record 1 private text 0.70 0 0.30 0 0
Record 2 private text 0.20 0.80 0 0 0
Record 3 private text 0 0.30 0.50 0.20 0
Record 4 private text 0.30 0 0 0.70 0
Record 5 private text 0 0.20 0.20 0 0.60
Public atom dictionary (public source)
CR Credit reporting
DC Debt collection
M Mortgage
CA Checking account
CC Credit card
• Each record is mapped to a K-sparse vector over the public atom dictionary. Example rows are illustrative.
• Each per-record vector is K-sparse ($K \leq 3$), nonnegative, and has total mass at most 1.
2 Aggregate semantic sketch
Sum the per-record vectors.
$C(D) = \sum_{i=1}^{n} \phi(x_i)$
Aggregate semantic mass over 5 illustrative records.
CR 1.20
DC 1.30
M 1.00
CA 0.90
CC 0.60
Compact vector
CR 1.20 DC 1.30 M 1.00 CA 0.90 CC 0.60
Ordered by decreasing mass.
3 DP plan release
Release a differentially private plan.
Input Aggregate Sketch
CR 1.20 DC 1.30 M 1.00 CA 0.90 CC 0.60
DP mechanism
Path A:
Full-vector perturbation
+
top-L atom selection
1) Perturb all coordinates.
CR 1.20 DC 1.30 M 1.00 CA 0.90 CC 0.60
+ noise
2) Select top-L atoms from perturbed masses
CR 1.29 DC 1.35 M 0.93 CA 0.92 CC 0.58
3) Release
DC 1.35 CR 1.29 M 0.93
Path B:
Randomized support selection
+
DP selected-value release
1) Select L atoms via EM
DC 1.30 CR 1.20 M 1.00
+ noise
2) Perturb selected values
DC 1.37 CR 1.14 M 0.98
3) Release
DC 1.37 CR 1.14 M 0.98
Released DP plan (top–3 admitted atoms)
Atom (admitted) Support bin
Debt collection High
Credit reporting High
Mortgage Medium
Privacy boundary
4 Decoder contract and verifier
Consumes only the DP plan.
Atom (admitted) Support bin
Debt collection High
Credit reporting High
Mortgage Medium
Uses only admitted atoms
Uses released bins
Rejects unsupported claims
5 LLM verbalization
Uses only the DP plan and public instructions.
LLM
Public instructions (e.g., summarization template, style)
6 Aggregate summary
Key themes
• Debt collection and credit reporting have high released support.
• Mortgage has medium released support.